

Number, units and estimation - Set 2 - Solutions

Specification M1.1, M1.2, M2.1-M2.14.

Answer key

Question	Answer	Topic
1	B	Number, units and estimation
2	D	Number, units and estimation
3	F	Number, units and estimation
4	A	Number, units and estimation
5	A	Number, units and estimation
6	A	Number, units and estimation
7	A	Number, units and estimation
8	H	Number, units and estimation
9	H	Number, units and estimation

Worked solutions

Question 1 NUM-16 · Number, units and estimation · spec M1.1

Water flows through a pipe at 1.2 litres per second. How long does it take to fill a tank of volume 3.6 m^3 ? ($1 \text{ m}^3 = 1000 \text{ litres}$.)

Answer: B

3600 litres at 1.2 litres/s takes $3000 \text{ s} = 50 \text{ minutes}$.

Question 2 NUM-17 · Number, units and estimation · spec M2.13

$a = 5.0$ and $b = 2.0$, each correct to 2 significant figures. Find the lower bound of $\frac{a}{b}$.

Answer: D

Smallest numerator over largest denominator: $\frac{4.95}{2.05}$.

Question 3 NUM-18 · Number, units and estimation · spec M2.3

What is the smallest positive integer that is divisible by every integer from 1 to 10?

Answer: F

$$\text{lcm}(1, \dots, 10) = 2^3 \cdot 3^2 \cdot 5 \cdot 7 = 2520.$$

Question 4 NUM-19 · Number, units and estimation · spec M2.9

Which of the following fractions has a terminating decimal expansion?

Answer: A

A fraction in lowest terms terminates iff its denominator has no prime factors other than 2 and 5: $40 = 2^3 \cdot 5$.

Question 5 NUM-20 · Number, units and estimation · spec M2.11

Find the exact value of $(\sqrt{5} - 1)^2 + (\sqrt{5} + 1)^2$.

Answer: A

$$(6 - 2\sqrt{5}) + (6 + 2\sqrt{5}) = 12.$$

Question 6 NUM-02 · Number, units and estimation · spec M2.3

The lowest common multiple of 12, 18 and n is 180. Which of the following could be n ?

Answer: A

$\text{lcm}(12, 18) = 36 = 2^2 \cdot 3^2$ and $180 = 2^2 \cdot 3^2 \cdot 5$, so n must be a divisor of 180 that is a multiple of 5. Only 20 qualifies: 24, 8 and 27 have no factor 5, and 50 and 40 do not divide 180.

Question 7 NUM-09 · Number, units and estimation · spec M2.3

How many positive divisors does 360 have?

Answer: A

$$360 = 2^3 \cdot 3^2 \cdot 5, \text{ so the number of divisors is } (3 + 1)(2 + 1)(1 + 1) = 24.$$

Question 8 NUM-12 · Number, units and estimation · spec M2.3

n is a positive integer such that n , $n + 2$ and $n + 4$ are all prime. Which of the following statements is/are true?

- I There is exactly one such value of n .
- II n must be odd.
- III One of n , $n + 2$, $n + 4$ must be divisible by 3.

Answer: H

Among any three numbers of the form n , $n + 2$, $n + 4$ one is a multiple of 3 (they cover all residues mod 3) - III. So one of them must equal 3 itself for all to be prime, giving $n = 3$ only (3, 5, 7) - I. And 3 is odd - II.

Question 9 X2-18 · Number, units and estimation · spec M2.3

n is a positive integer greater than 1. Which of the following statements is/are true?

- I If n is not prime, then n has a prime factor at most $\sqrt{n}$.
- II If n has no prime factor at most $\sqrt{n}$, then n is prime.
- III If n is prime, then n has no factor between 1 and n exclusive.

Answer: H

If $n = ab$ with $1 < a \leq b$ then $a \leq \sqrt{n}$ and a has a prime factor $\leq \sqrt{n}$ (I). II is the contrapositive of I. III is the definition of prime.